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CE F231
Fluid Mechanics

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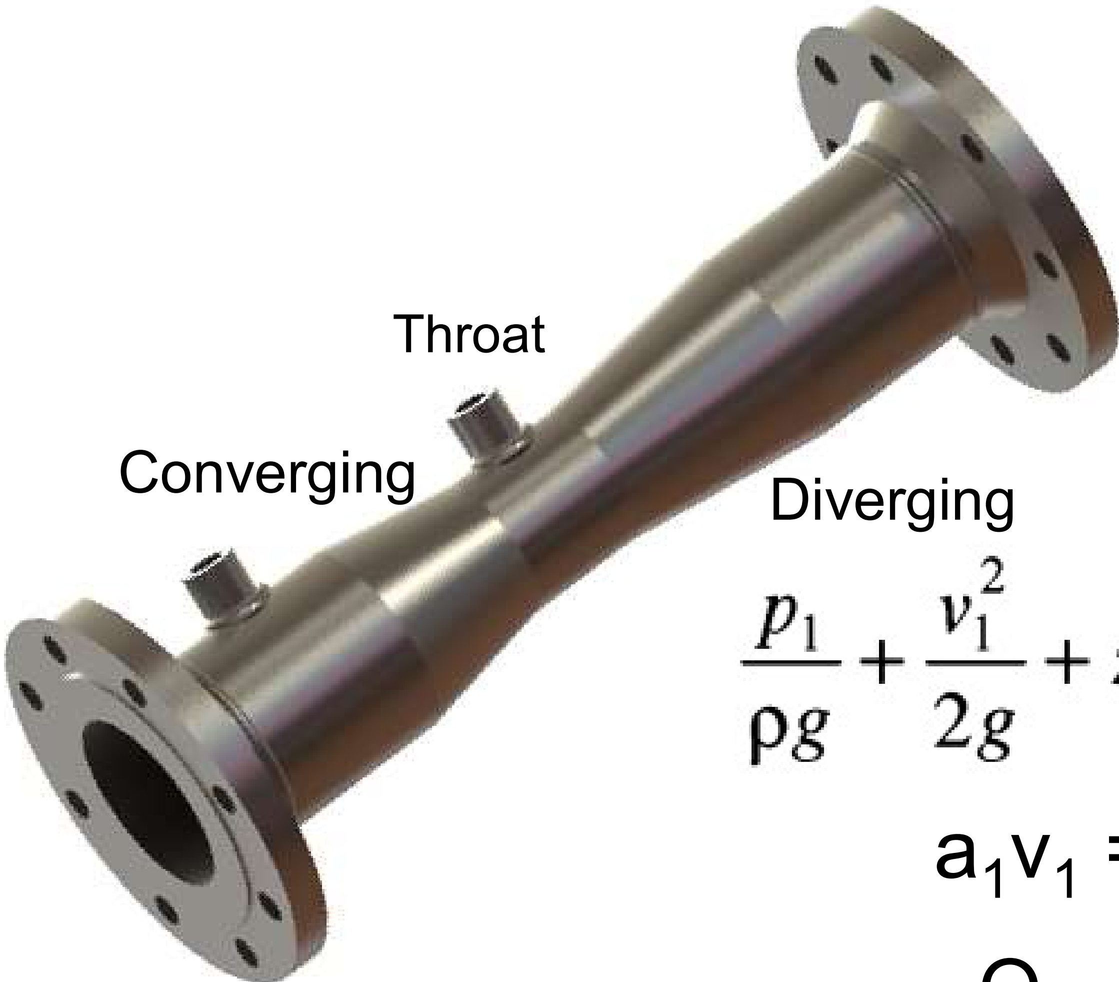
Lecture 23: Fluid Dynamics

Recap: Venturi meter

- Used for measuring volume flow rate of fluid through pipe



$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + h_L$$

$$a_1 v_1 = a_2 v_2 = Q_{theo}$$

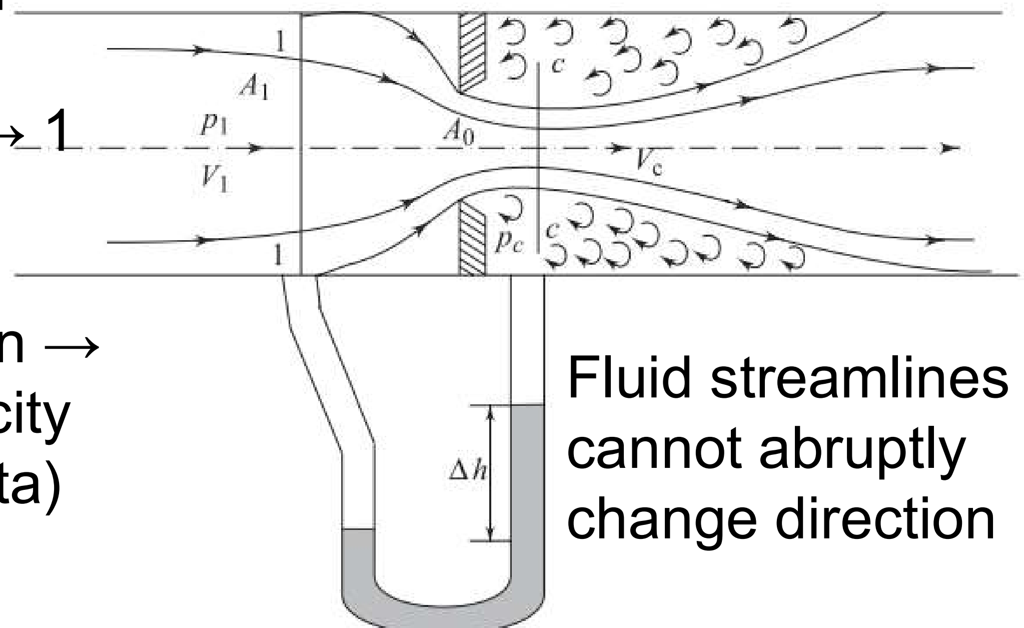
$$Q_{act} = C_d \times Q_{theo}$$

https://chemicalengineeringworld.com/venturi-meter-versus-orifice-meter/

https://www.mechanicalbooster.com/2016/06/venturimeter-principle-construction-working-diagram.html

Orifice meter

- ➢ Used for measuring volume flow rate of fluid through pipe
- ➢ Simpler and cheaper than venturi meter
- ➢ Thin circular plate with sharp edged concentric circular hole
- ➢ Orifice diameter is 0.5 x pipe diameter
- ➢ Upstream pressure tapping location → 1 to 2 x pipe diameter
- ➢ Downstream pressure tapping location → 0.5 x orifice diameter (maximum velocity or minimum pressure at vena contracta)

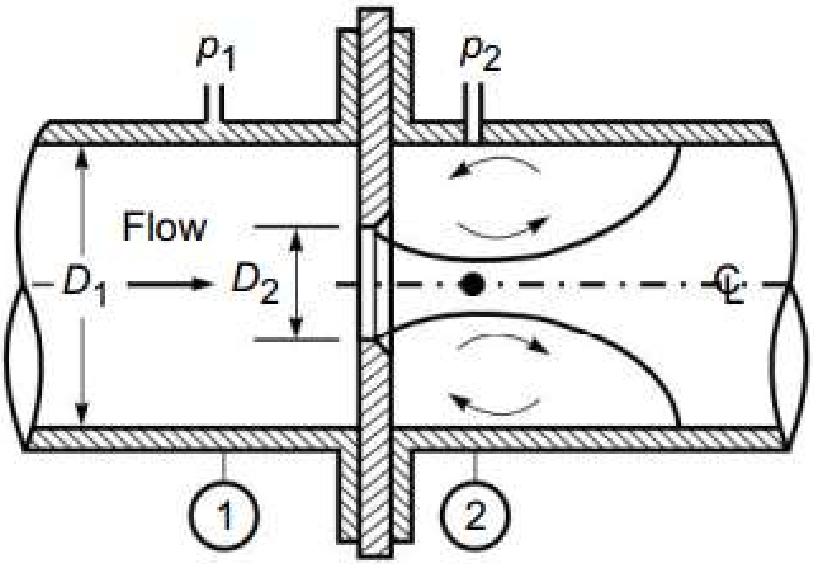


Fluid streamlines cannot abruptly change direction

https://www.armstronginternational.com/products-and-systems/flow-measurement/armstrong-orifice-plate https://www.pipingengineer.org/orifice-flow-meter/

Orifice meter

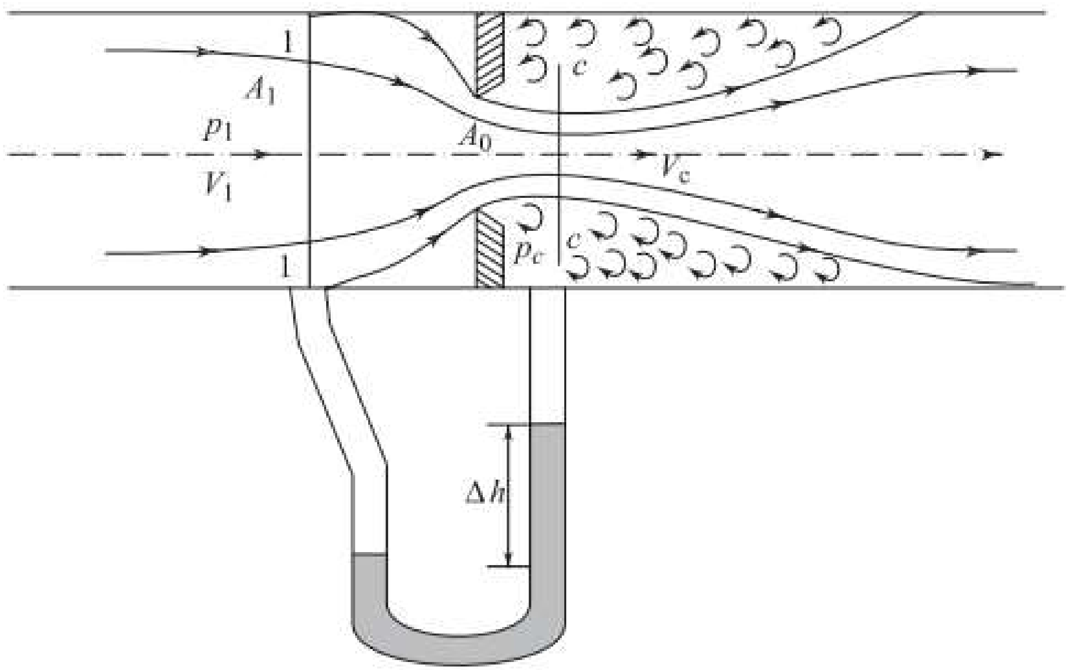
An orifice meter with orifice diameter 15 cm is inserted in a pipe of 30 cm diameter. The pressure gauges fitted upstream and downstream of the orifice meter give readings of 14.715 N/cm^2 and 9.81 N/cm^2 respectively. Find the rate of flow of water through the pipe in litres/s. Take $C_d = 0.6$. [**Ans.** 108.434 lit/s]
$C_v = 0.98$



$$Q = C_d A_2 \frac{\sqrt{2g\Delta H}}{\sqrt{1 - C_c^2 (D_2/D_1)^4}} \qquad (13.7)$$

where $\Delta H = \dfrac{(p_1 - p_2)}{\gamma}$ = difference in pressure heads upstream and downstream of the orifice plate.

Orifice meter



$$Q = C_c A_0 C_v \sqrt{\frac{2g(\rho_m/\rho - 1)}{(1 - C_c^2 A_0^2/A_1^2)} \Delta h}$$

where Δh is the difference in liquid levels in the manometer and ρ_m is the density of the manometric liquid.